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OCAD Summer Internship Project

Advanced Cross-Platform GNSS API for Android and iOS

Overview

This project aims to investigate, design, and prototype a more advanced and configurable GNSS (Global Navigation Satellite System) data and processing framework for Android and iOS platforms. The goal is to provide significantly more control, visibility, and extensibility than the default mobile location APIs currently expose.

Modern mobile operating systems provide only limited access to GNSS information, typically abstracting away satellite data, signal properties, correction services, and low-level measurements. This project will explore how to expose richer GNSS functionality through a unified, self-contained cross-platform module.

The module should support advanced GNSS use cases including:

- Access to detailed satellite and signal metadata
- Configurable GNSS update behaviour and filtering
- Integration with external Bluetooth GNSS receivers
- Access to raw measurements and signal processing data where available
- A consistent API across Android and iOS

The scope is intentionally open-ended to encourage research, experimentation, and discovery. Any progress made in the areas below would provide value to the wider platform and future development work.

Project Objectives

Primary Objectives

- Research GNSS capabilities and limitations on Android and iOS
- Design a unified, extensible GNSS API abstraction
- Implement a self-contained cross-platform module
- Expose more detailed GNSS data than standard mobile APIs provide
- Improve configurability and control over GNSS behaviour

Secondary / Stretch Objectives

- Bluetooth external GNSS receiver integration
- Raw GNSS measurement access
- Signal quality analysis and diagnostics
- Configurable filtering and signal processing
- Performance and accuracy benchmarking

Desktop and Future Platform Considerations

While the primary focus is use with mobile platforms, it may also be beneficial to briefly investigate future extensibility toward desktop platforms. Modern desktop operating systems also expose GNSS and location APIs with varying levels of capability and hardware support. A modular and extensible architecture could help support future experimentation or expansion beyond mobile platforms.

Background

Current mobile location APIs are designed primarily for consumer location services and therefore expose only limited information such as:

- Latitude / longitude
- Accuracy estimates
- Speed and heading
- Basic satellite counts

However, modern Android devices and external GNSS receivers can provide significantly more information, including:

- Satellite constellation data (GPS, Galileo, GLONASS, BeiDou, etc.)
- Signal strength and carrier properties
- Raw pseudorange measurements
- Carrier phase measurements
- Multi-frequency support

A richer abstraction layer would allow applications to:

- Improve positioning accuracy
- Support surveying and mapping workflows
- Analyse signal quality

- Integrate professional-grade GNSS hardware
- Experiment with advanced positioning techniques

Proposed Areas of Investigation

1. Native GNSS Capability Research

Investigate what data and APIs are available on:

- Android GNSS APIs
- iOS Core Location and related frameworks
- Device-specific limitations
- Background execution constraints

2. Cross-Platform API Design

Design a clean and extensible API surface that abstracts platform differences while still exposing advanced functionality.

Possible Android GNSS API Features

- Satellite visibility data
- Signal quality metrics
- Constellation information
- Raw measurement access
- Event-driven GNSS updates

Unified platform API Features

- Configurable update rates
- Diagnostic information
- Logging and telemetry

3. Bluetooth GNSS Receiver Integration

Research and prototype communication with external GNSS receivers over Bluetooth.

Potential Features

- NMEA stream parsing
- Vendor-specific protocol support
- Receiver discovery and pairing
- Real-time position ingestion
- Multi-device compatibility

5. Signal Processing and Diagnostics

Explore exposing lower-level signal information and configurable processing options.

Potential Features

- Signal filtering
- Noise analysis
- Multipath detection
- Measurement smoothing
- Accuracy estimation

Technical Considerations

Proposed Technology Stack

- **Language:** C#
- **Platforms:** Android and iOS
- **Frameworks:** .NET / MAUI

Architecture Goals

- Self-contained module
- Minimal platform-specific code exposure
- Extensible plugin-style design
- Consistent cross-platform API
- Testable and maintainable structure

Potential Challenges

- Platform limitations and API restrictions
- Inconsistent GNSS feature availability across devices
- iOS restrictions on low-level GNSS access
- Bluetooth compatibility differences
- Battery and performance considerations
- Hardware dependency for testing advanced features

The mobile platforms may expose substantially different GNSS capabilities and restrictions, particularly around low-level measurements, background operation, Bluetooth integration, and raw signal access. In practice, some advanced functionality may only be achievable on one platform.

The project should therefore prioritise:

- Providing the best possible native functionality on each platform
- Maintaining a clean and consistent API shape where practical
- Clearly documenting platform-specific capabilities and limitations

A useful platform-specific feature is still considered valuable even if an equivalent implementation is not currently possible on the other operating system.

Additional Real-World Use Cases

The project should also consider practical operational scenarios encountered when using professional or semi-professional external GNSS hardware in real mobile workflows.

External Bluetooth GNSS Reliability and Monitoring

On Android, external GNSS receivers are often integrated through third-party applications (Bluetooth GNSS). In these workflows, connectivity failures can occur without the user immediately noticing, resulting in a silent degradation in positioning accuracy.

Potential areas of investigation could include:

- Detection of dropped Bluetooth GNSS receiver connections
- Detection of loss of RTK correction streams or internet connectivity
- User-visible warnings and diagnostic status reporting
- Automatic reconnection and recovery strategies
- Confidence/health indicators showing whether high-accuracy positioning is currently active
- Monitoring whether the device is currently using internal or external GNSS sources

Expected Outcomes

By the end of the internship, expected outputs may include:

- Research findings and technical documentation
- A prototype GNSS abstraction and processing layer
- Cross-platform API design
- Working proof-of-concept implementations
- Bluetooth GNSS integration experiments
- Recommendations for future production development

The project is exploratory in nature, and success is measured not only by implementation progress but also by the quality of research, findings, and architectural direction established for future work.

References and Related Material

OCAD Desktop Real Time GPS

- WIKI
- NMEA Parsing

External GNSS Interface Applications

- Bluetooth GNSS

Summary

This project explores the development of a modern, extensible, and configurable GNSS data and processing framework for mobile platforms, with a focus on exposing and utilising advanced positioning capabilities not typically available through standard mobile location APIs.

The work combines research, systems design, native platform integration, and experimental prototyping, with opportunities to investigate external GNSS hardware and signal analysis techniques.

The open-ended nature of the project allows flexibility to focus on the most promising or interesting areas discovered during the internship.